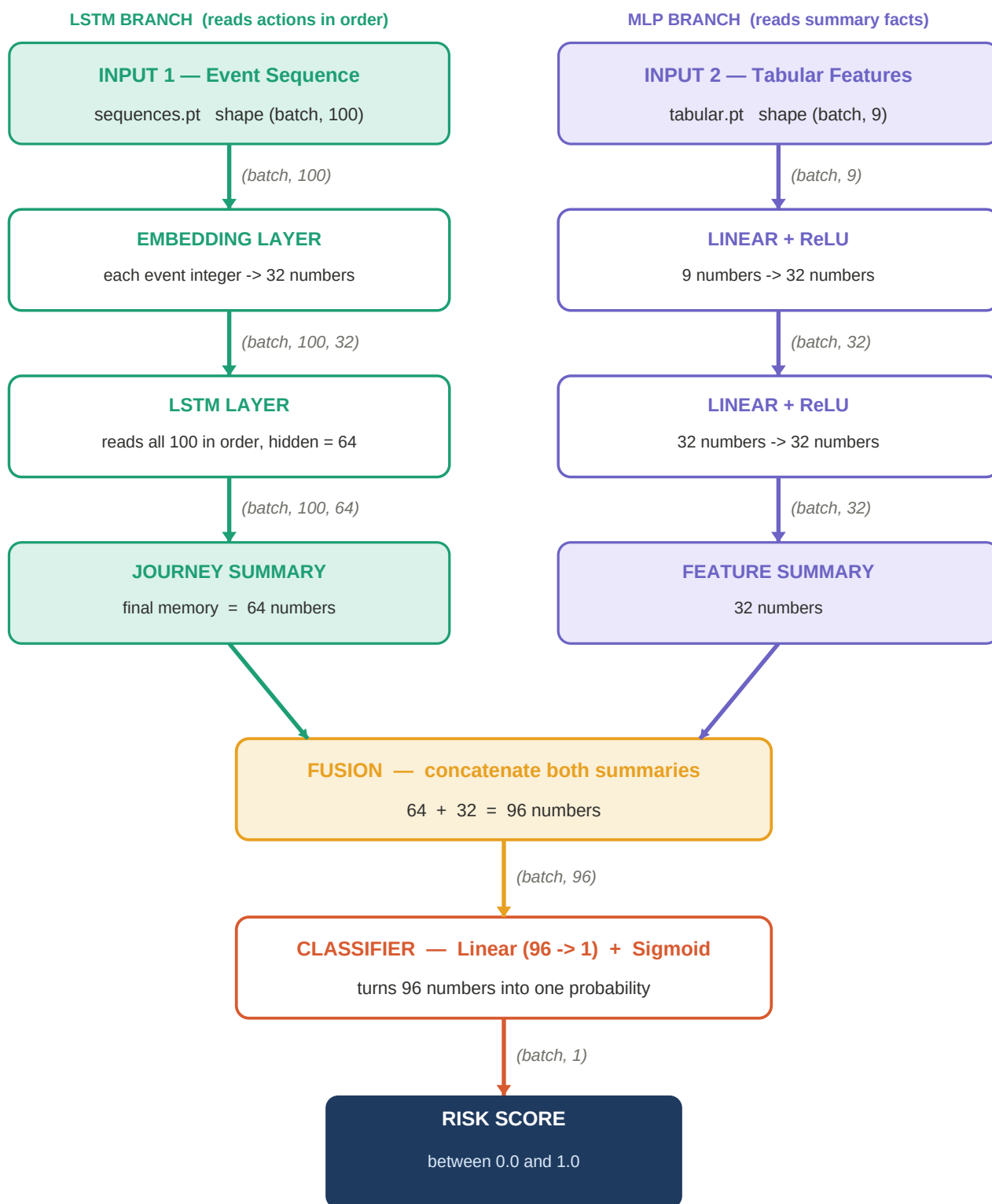


PulseAI — Model Architecture

Two branches read the customer, then merge into one risk score



In plain words:

The left branch summarises WHAT the customer did (in order). The right branch summarises FACTS about them. We glue both summaries together and turn them into a single churn-risk score. Page 2 has the exact layer details.

Layer-by-Layer Detail

Follow the data shape through every layer — this is your interview walkthrough

BRANCH 1 — Event Sequence (LSTM)

1. Input

(batch, 100)

100 padded event integers per customer. 0 means padding (nothing happened).



2. nn.Embedding

(batch, 100, 32)

Each integer becomes a learnable 32-number vector. padding_idx=0 keeps padding as zeros.



3. nn.LSTM

(batch, 100, 64)

Reads the 100 vectors in order, updating a memory. hidden_size=64, batch_first=True.



4. Last hidden

(batch, 64)

We keep only the FINAL memory — a 64-number summary of the whole journey.

BRANCH 2 — Tabular Features (MLP)

1. Input

(batch, 9)

9 summary features: failure rate, tenure, plan, tickets, open rate, cancel visit, etc.



2. Linear + ReLU

(batch, 32)

Dense layer expands 9 -> 32. ReLU adds non-linearity so it can learn complex patterns.



3. Linear + ReLU

(batch, 32)

Second dense layer, 32 -> 32, for a deeper representation of the facts.

FUSION + CLASSIFIER

Concatenate: `torch.cat([seq64, tab32], dim=1) -> (batch, 96)`

Linear: `nn.Linear(96, 1) -> (batch, 1)`

Sigmoid: squashes to a probability between 0.0 and 1.0 = RISK SCORE